Matt J Bryan BA MEng (Cantab) MIET

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Education

University of Cambridge - Magdalene College — Oct 2024 –

→ PhD, Engineering - fully funded by Bose (MA, USA)

- **Project:** ***Hybrid physics-ML models for active noise control in automotive systems***
- **Research:** Incorporating underlying physics into machine learning (ML) architectures to improve performance in active noise control (ANC), which requires models to be *both* accurate and fast.
- This combines expertise in physical domains (nonlinear dynamics, vibration, acoustics) with control theory and data-driven ML methods to solve complex problems e.g. real-time prediction of sound through a vehicle.
- Tailored physics-ML architectures show promise and benefits compared to pure physics and data methods.
- **Supervisors:** Dr Tore Butlin (CUED) and Dr Ole Nielsen (Bose/CUED).

University of Cambridge - Magdalene College — 2020 – 2024

→ BA MEng, Engineering (Mechanical & Control)

- **Grades:** 1st Class (BA), Distinction (MEng). 1st in all graded years.
- Two general years, before dual specialisation in Mechanical and Control Engineering in the third year. Advanced study of vibration, nonlinear dynamics, control, and acoustics in the fourth year.
- **MEng Project:** *'Pushing the bounds of energy harvesting'* - 1st Class.
  Investigated circumventing the 'mass bound' for general energy harvesting systems. Developed a design from theory to practice with a novel dual translation and rotationally excited piezoelectric harvester.

Teaching

→ **Supervisor** - Magdalene/Clare College, Cambridge — Oct 2024 –

- Passionate about teaching across all levels of engineering with c. 30 students p.a. In high demand as a supervisor with strong results - my IA students performed 10% higher than the cohort average last year.
- Supervising students for 1st year Mechanics at Magdalene, 2nd year at Clare, and 3rd year advanced Dynamics and Vibration courses. Plus, a number of research projects for the 'Bicycle Design' course.
- Currently completing an **AFHEA** teaching qualification through selective University Scheme.

→ **Admissions Interviewer** - Magdalene/Emmanuel College, Cambridge — Dec 2024 –

- One of Magdalene's youngest ever interviewers (at 22 years-old), designing technical questions for and conducting 30+ engineering undergraduate admission interviews p.a. Contributing to decision process.
- Working closely with Dr Roebuck & Dr Ravenscroft at Magdalene. Also invited to interview for Emmanuel.

Industrial Experience

→ **R&D Intern** - Siemens Healthineers Magnet Technology, Oxford — 2022, 2023

- Improved imaging and patient diagnostic outcomes during two summer placements developing superconducting magnets for MRI systems - developed a novel vibration measurement technique in a strong magnetic field, and used finite-element simulation of undesirable field-induced vibration to inform design.
- Selected for Siemens' Cambridge Sponsorship Scheme run by Prof. D Cardwell following a highly competitive interview process (1 of 2 students across physical sciences, engineering, and computer science).

Publications

[1] PATENT - **Novel machine learning architecture for active noise control in nonlinear systems**

- From my PhD work. Application filed with Cambridge Enterprise and Bose in Feb 2026.

[2] PAPER - **Physics-informed machine learning models for automotive active noise control**

- Accepted conference paper on novel architecture, to be presented at the International Conference on Noise and Vibration Engineering (ISMA), Leuven BE, September 2026.

NB: Bose studentship includes an intellectual property agreement, so patents are required before publications.

Academic Activities

- **Secondment:** Bose Research, Boston MA, Summer 2026 (UPCOMING).
  Planned work on Physics-Informed ML in real automotive systems, forming part of my PhD thesis.
- **Convener**: Dynamics and Vibration Research Group (DVRG) Seminar Series, 2025 - present
  Scheduling and chairing weekly talks with internal and invited speakers across dynamics, vibration, and ML.
- **Workshop Talk:** *'Hybrid models for Active Noise Reduction'*
  Institute of Physics (IOP) Physics-Informed Machine Learning Workshop, London, October 2025.

Awards

→ **PhD Studentship** - Bose at the University of Cambridge — OCT 2024 -

- Studentship fully funded by Bose Corporation for 4 years of doctoral study including research budget and secondment at Bose Research in Boston MA. Awarded following a competitive interview process.

→ **Best Poster** - CUED Division C Graduate Conference 2025 — MAY 2025

- Awarded for poster and presentation on *'ML for Active Noise Control'*.

→ **College Scholarships & Prizes** - Magdalene College, Cambridge — OCT 2020 -

- Current **Senior Scholar** for a Distinction in my Master's. Awarded an academic scholarship every year of my undergraduate degree for First Class results, including Christopherson and Foo Sun Lau prizes.

→ **Undergraduate Scholarship** - Institute of Mechanical Engineers — OCT 2020 - JUN 2024

- Academic scholarship awarded for my MEng following a competitive application process (1 of 6 nationally).

Extracurricular

Memberships: Institution of Engineering and Technology (MIET), Institution of Mechanical Engineers (AMIMECHE), Cambridge Philosophical Society (FCPS).

Volunteering: Cambridge University Cycling Club Committee, various roles inc. General Secretary, Welfare Officer (2021 -), CUED Outreach, Cambridge Slipstream Mentoring Scheme.

Hobbies: Cycling (road and mountain-biking) → 2× Half Blue. Choral singing → Magdalene (*Choral Scholar*, Garrett Music Prize, Music Soc Treasurer) & Queens' College Choirs, Portsmouth Cathedral Choir.

Referees

Dr Tore Butlin — `tb267-at-cam.ac.uk`
- University Associate Professor, CUED (Principal PhD Supervisor)

Dr Richard Roebuck — `rlr20-at-cam.ac.uk`
- Director of Studies in Engineering (Part I), Magdalene College, Cambridge